

Solution

1) Calculons d'abord le t.f. de $2^{-|t|} = 2^{-t} 1(t) + 2^t 1(-t)$

$$2^{-t} 1(t) = e^{-t \cdot \ln 2} 1(t)$$

$$\int_0^{+\infty} e^{-t(\ln 2)} e^{-j2\pi f t} dt = \frac{1}{\ln 2 + j2\pi f}$$

$$2^{-|t|} \longleftrightarrow \frac{1}{\ln 2 + j2\pi f} + \frac{1}{\ln 2 - j2\pi f} = \frac{2 \ln 2}{(\ln 2)^2 + 4\pi^2 f^2}$$

$$0.5 2^{-\frac{|t-3|}{2}} \longleftrightarrow \frac{2 \ln 2}{(\ln 2)^2 + 16\pi^2 f^2} \cdot e^{-j2\pi 3f}$$

$\uparrow$ constante
 $\uparrow$ cambio de fase

2) $x(t)$ è la periodizzazione di $x_1(t) = 2^t \cdot 1(-t)$.

Dunque $X\left(\frac{k}{T}\right) = X_1\left(\frac{k}{T}\right)$ (Periodizzazione $\rightarrow$ componenti).

$$\text{ove } x(t) = X(0) = X_1(0) = \text{ove } x_1(t)$$

$$= \int_{-\infty}^0 2^t dt = \frac{1}{\ln 2}$$

$$(2^t = e^{t \ln 2})$$

$$3) \quad m_e = \frac{1}{2} \quad M_e = \frac{1}{2} \quad \sigma_e^2 = M_e - m_e^2 = \frac{1}{4}$$

$$R_e(f) = \sigma_e^2 T + m_e^2 \sum_k \delta_{12}\left(f - \frac{k}{T}\right)$$

$v(k)$: analog filter interpolator can replace
impulsive $\frac{1}{T} \text{sinc}^2 \frac{t}{T}$, $H(f) = \text{sampling } f \cdot T$

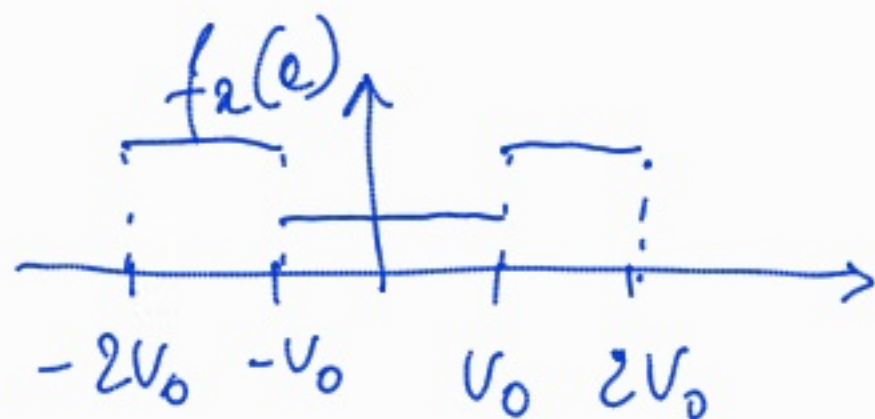
$$\bar{R}_v(f) = R_e(f) \cdot |H(f)|^2 = \left(\sigma_e^2 T + m_e^2 \sum_k \delta_{12}\left(f - \frac{k}{T}\right) \right) \cdot \text{sampling}^2 f T$$

$$= \sigma_e^2 T \text{sampling}^2 f T + m_e^2 \delta_{12}(f)$$

$$\bar{M}_v = \sigma_e^2 \cdot T \cdot 2 \frac{1}{3T} + m_e^2$$



4)



$$\Delta = \frac{4V_0}{16}$$

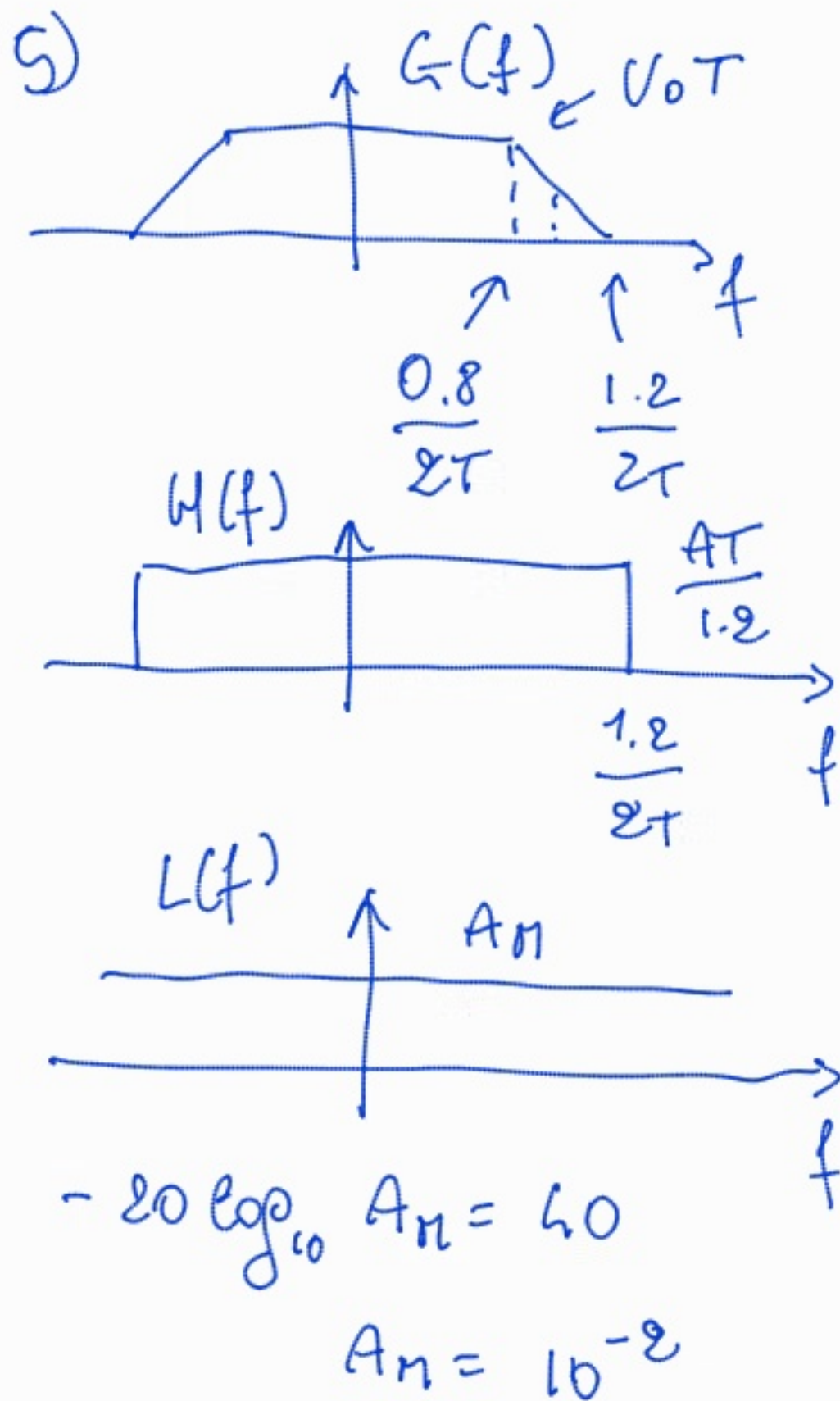
Dato che $f_x(e)$ è costante all'interno degli intervalli Δ di quantizzazione e nullo per $|e| > 2V_0$, essendo la regione preimage uguale a $[-2V_0, 2V_0]$, la potenza dell'errore di quantizzazione è esattamente

$$M_e = \frac{\Delta^2}{12} = \frac{V_0^2}{192}$$

$$M_x = \int_{-V_0}^{V_0} \frac{1}{6V_0} e^2 de + 2 \int_{V_0}^{2V_0} \frac{1}{3V_0} e^2 de = \frac{5V_0^2}{3}$$

$$\frac{M_x}{M_e} = 320$$

$$10 \log_{10} \frac{M_x}{M_e} \approx 25.05 \text{ dB}$$



$$C(f) = G(f) W(f) L(f)$$

$$= \frac{AT}{1.2} A_n G(f)$$

$$c(f) = \frac{AT}{1.2} A_n V_0 2 \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.2t}{T}\right)$$

dw
Nyquist
↓

$$V'_0 = c(0) = \frac{AT}{1.2} A_n V_0$$

$$\sigma_n^2 = R_0 E_h = R_0 \left(\frac{AT}{1.2}\right)^2 \cdot \frac{1.2}{T}$$

$$P_e = \frac{2.3-2}{3} Q\left(\frac{V'_0}{2\sigma_n}\right)$$